QCM A · read and check a representation

Orthonormal coordinate system. After S67; QCM B after S70.

For each question, only one proposal is accurate.
consult the correction.

1. $d : (x; y; z) = (1; 0; 2) + t(2; -1; 0)$ admits as director:

A. $(1; 0; 2)$ B. $(2; -1; 0)$

C. $(2; 1; 0)$ D. $(0; 0; 2)$

2. Which of these points belongs to d ?

A. $(3; 1; 2)$ B. $(1; 0; 0)$

C. $(2; -1; 0)$ D. $(3; -1; 2)$

3. A normal vector on the plane $x + 2y - 2z = 3$ is:

A. $(1; 2; -2)$ B. $(1; 2; 3)$

C. $(3; 0; 0)$ D. $(-2; 1; 0)$

4. To check H Projected M on $\mathcal{P}$, it must:

A. Only $H \in \mathcal{P}$ B. Only MH small

C. $H \in \mathcal{P}$ and $\overrightarrow{MH}$ normal on the plan D. That $H = M$

5. Right $A + tu$ with $u \neq 0$ is also represented by:

A. $A + t\vec{0}$ B. $A + 2u + s(3u)$

C. $tA + u$ D. the only point $A + u$

MCQ B · intersections and special cases

For each question, only one proposal is accurate.
consult the correction.

6. The substitution of a line in a plane equation gives 0 $= 4$. The intersection is:

A. a right B. One point

C. the plan D. empty

7. If substitution gives 0 $= 0$ for any parameter:

A. the right is included in the plane B. there is no common ground

C. There is only one point in common D. The data are incorrect

8. To search for the intersection of $A + tu$ and $B + sv$We have to:

A. Impose $s = t$ B. Replace both parameters with 0

C. Solve for s, t Independent D. ignore the third co-ordinate

9. The intersection of $x + y = 1$ and $z = 2$ is:

A. a plan B. Right $(t; 1 - t; 2)$

C. the point $(1; 0; 2)$ D. empty

10. For $m = 0$, total $mx + y = 1$ is:

A. Empty B. a right

C. All space D. The plan $y = 1$

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Two straight lines which have non-colonial directors are necessarily stinging.

2. Plans $x + y + z = 1$ and $2x + 2y + 2z =$ Three are confused.

3. The proposed M on a plan is unique.